

Cyber Security

Introduction



https://techgirls.ece.vt.edu/slides/introduction_to_cybersecurity.html#39

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Introduction

- Crime committed using a computer and the internet to steal a person's identity or illegal imports or malicious programs. Cyber crime is an activity done using computers and the internet.
- Cyber security refers to the technologies and processes designed to protect computers, networks and data from unauthorized access and attacks delivered via the internet by cyber criminals. Though, cyber security is important for the network, data and application security.
- The objective of cyber security is to establish rules and measure to use against attacks over the internet.

Introduction

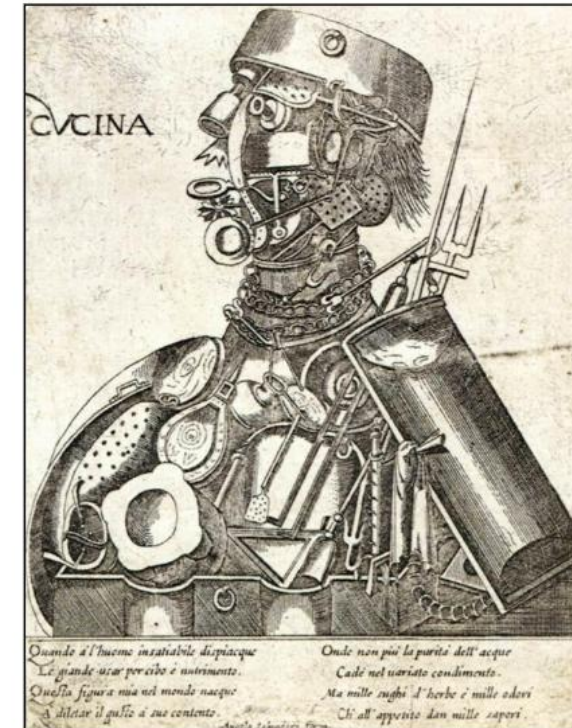
What is Cyberspace?

- Cyberspace is "the notional environment where communication over computer networks occurs." -- Wikipedia
- "Cyberspace" term coined by William Gibson - 1984

What is Cybersecurity?

"Computer security, also known as cyber security or IT security, is the protection of computer systems from the theft or damage to their hardware, software or information, as well as from disruption or misdirection of the services they provide." -- Gasser, Morrie (1988).

Includes *physical* security as well
as *online* security



Humani Victus Instrumenta: Ars Coquinaria, 1569, Copper engraving,

379 x 277 mm, Private collection

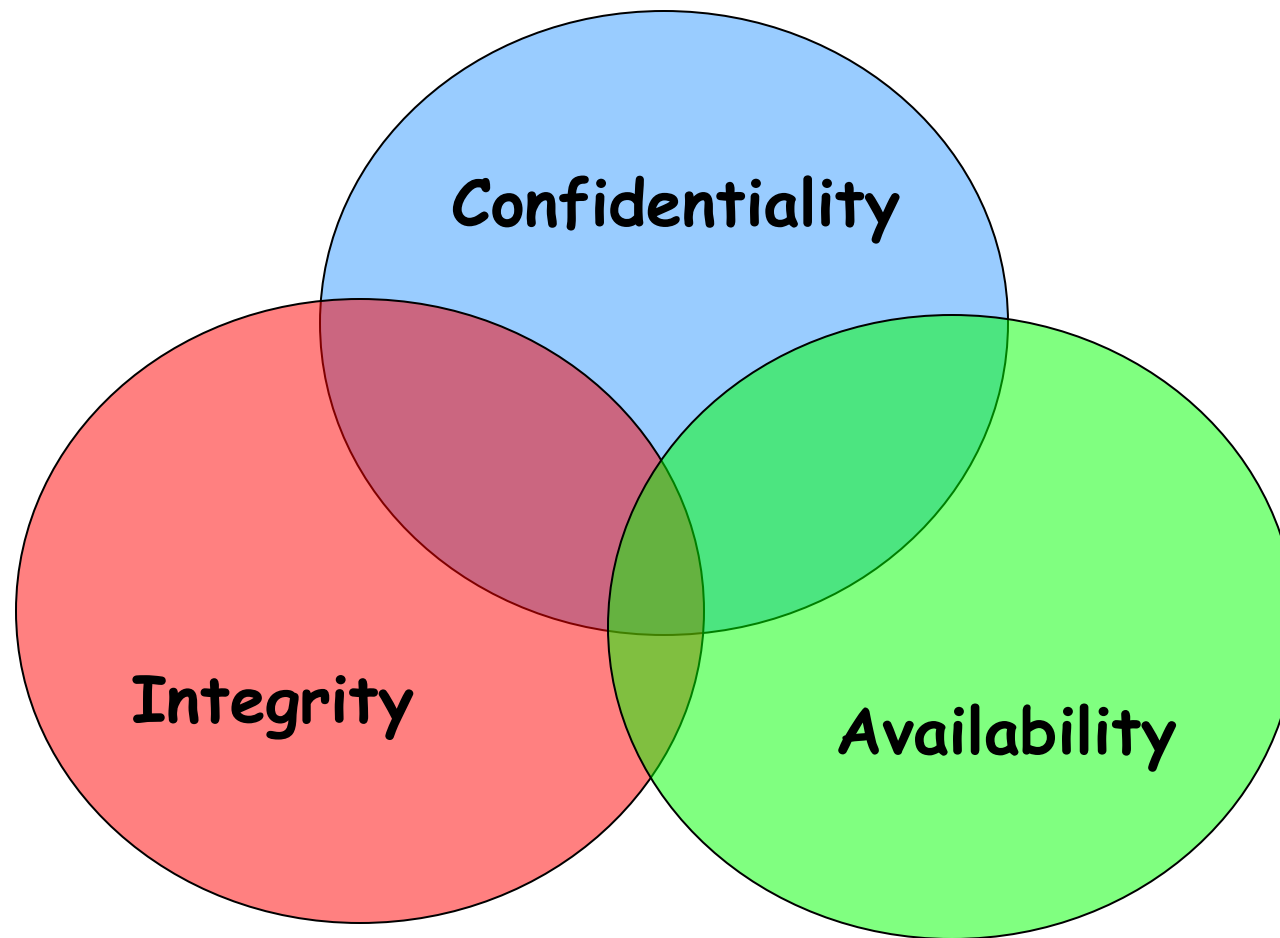
Outline

- Attacks, services and mechanisms
- Security attacks
- Security services
- Methods of Defense
- A model for Internetwork Security
- Internet standards and RFCs

Security?

- What does it refer to?
- Three Fundamentals
 - Confidentiality
 - Authorized users get entry
 - Inspection of information, printing of information and knowledge of resource existence
 - Availability
 - Legitimate user be able to access resources and service should be provided at anytime
 - Integrity
 - Changed in adequate way
 - Modification done by authorized people

Security Goals



Attack Process

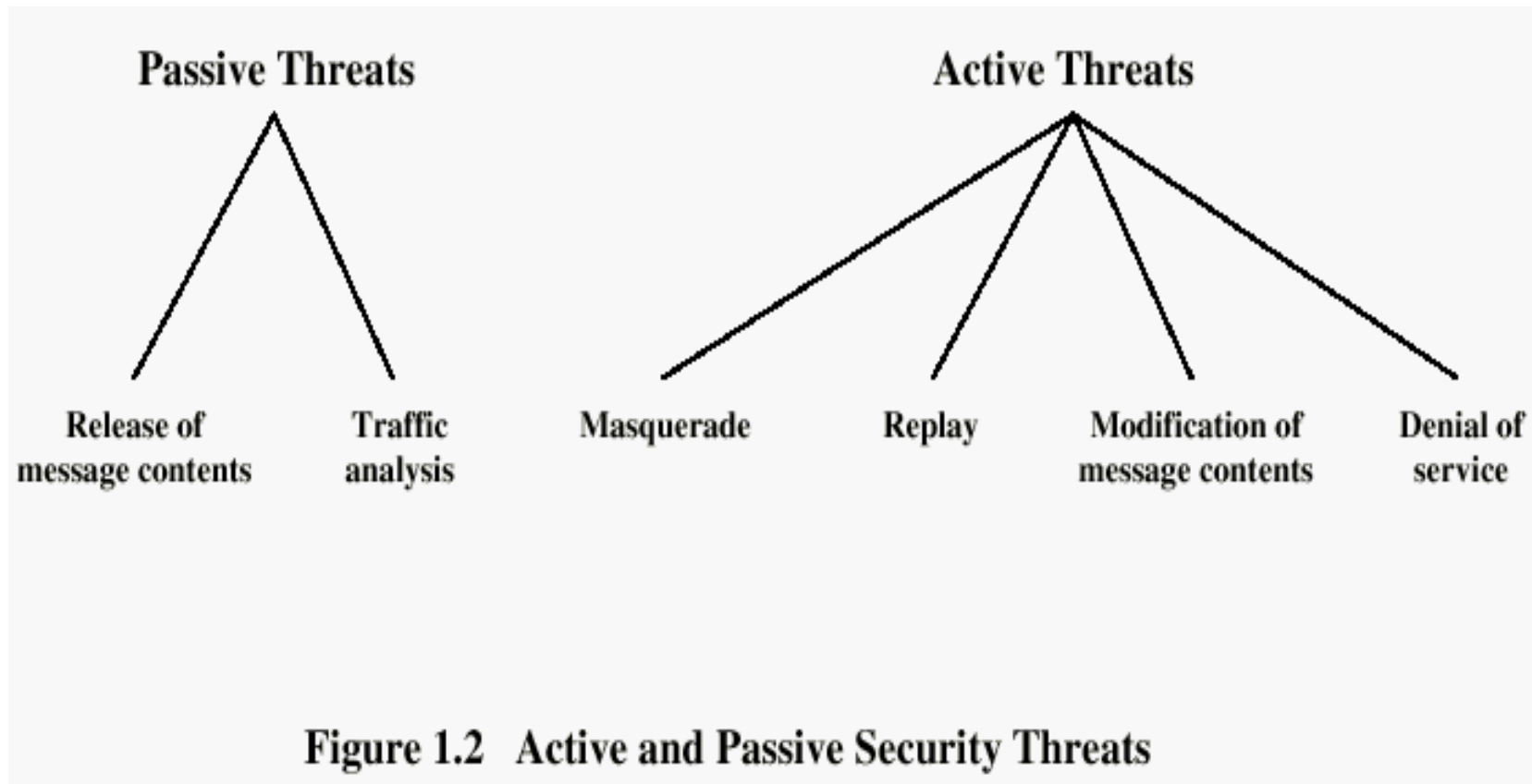
- Three stages called MOM
- Method
 - Ability, information, tools
- Opportunity
 - Time and Access
- Motive
 - Testing system reliability
 - Competition between attackers or testing their own skills
 - Breaking into well secured systems like law enforcement, government agencies
 - To gain popularity, financial gain, information gain
 - Just for fun
 - No motive at all

The OSI (open systems interconnection) security architecture

- **Security Attack:** Any action that compromises the security of information.
- **Security Mechanism:** A mechanism that is designed to detect, prevent, or recover from a security attack.
- **Security Service:** A service that enhances the security of data processing systems and information transfers. A security service makes use of one or more security mechanisms.

Security Attacks

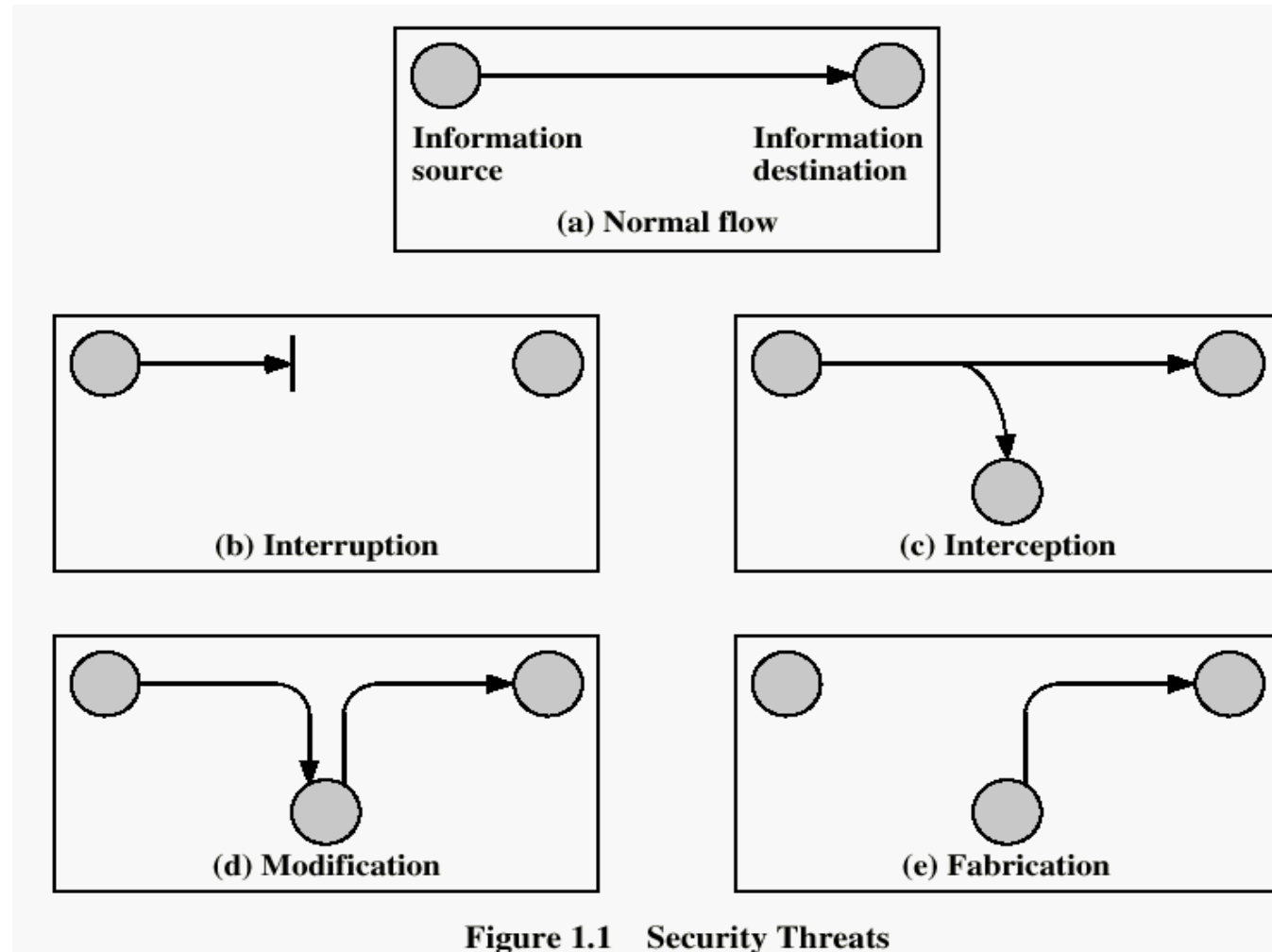
- **Passive Attack**
- Passive attacks are in the nature of eavesdropping on, or monitoring of, transmissions. The goal of the opponent is to obtain information that is being transmitted
 - attempts to learn or make use of information from the system but does not affect system resources
- **Active Attack**
 - attempts to alter system resources or affect their operation
 - Active attacks involve some modification of the data stream or the creation of a false stream and can be subdivided into four categories



Security Attacks

- **Interruption:** This is an attack on availability
- **Interception:** This is an attack on confidentiality
- **Modification:** This is an attack on integrity
- **Fabrication:** This is an attack on authenticity

Security Attacks



Security Services

- Processing or communication service that is provided by a system to give a specific kind of protection to system resources; security services implement security policies and are implemented by security mechanisms
 - Confidentiality (protect data from unauthorized disclosure)
 - Authentication (who created or sent the data)
 - Data Integrity (has not been altered)
 - Non-repudiation (the order is final) Non repudiation prevents either sender or receiver from denying a transmitted message. Thus, when a message is sent, the receiver can prove that the alleged sender in fact sent the message. Similarly, when a message is received, the sender can prove that the alleged receiver in fact received the message.
 - Access control (prevent misuse of resources)
 - Availability (property of a system or a system resource being accessible and usable upon demand by an authorized system entity)
 - Denial of Service Attacks

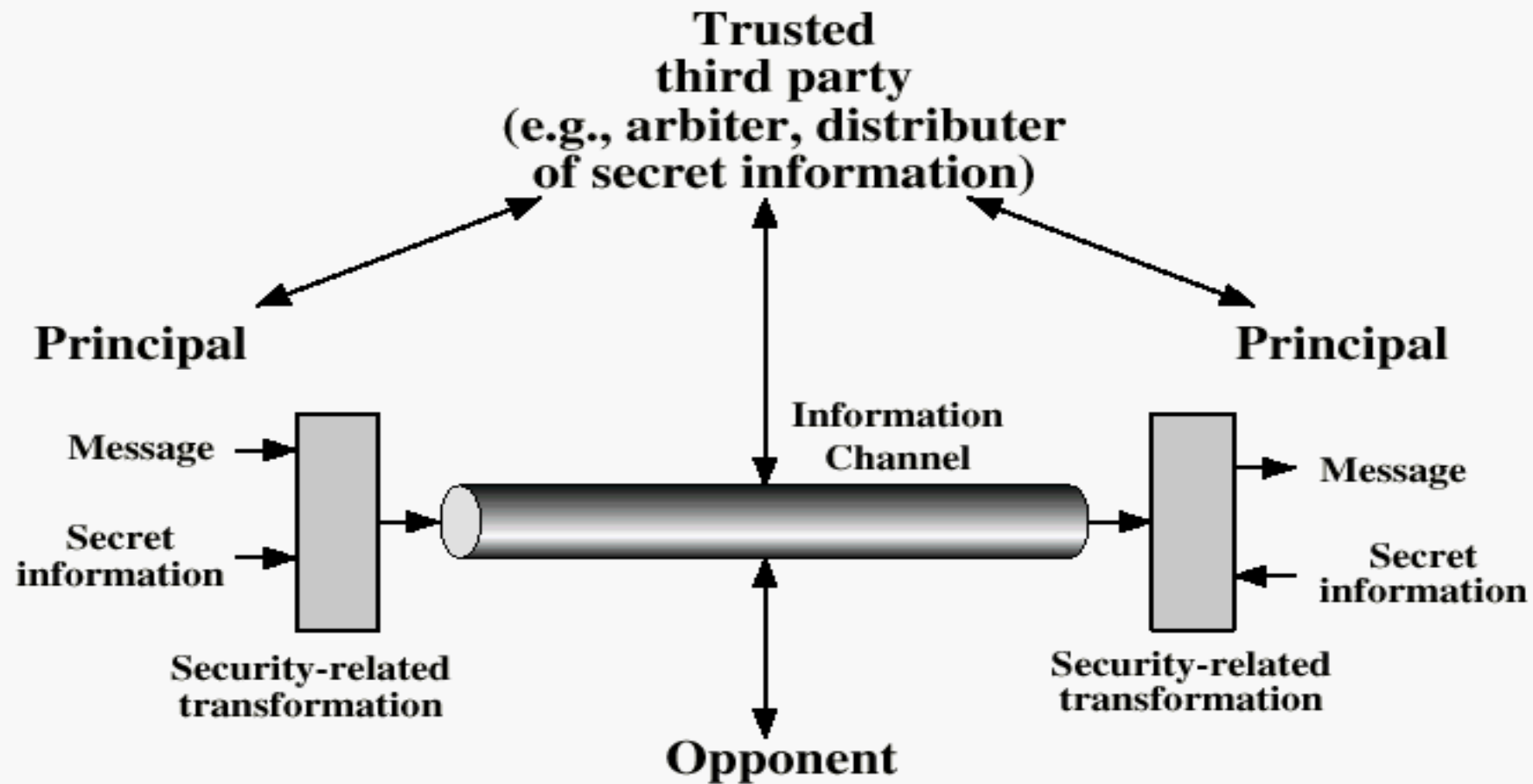


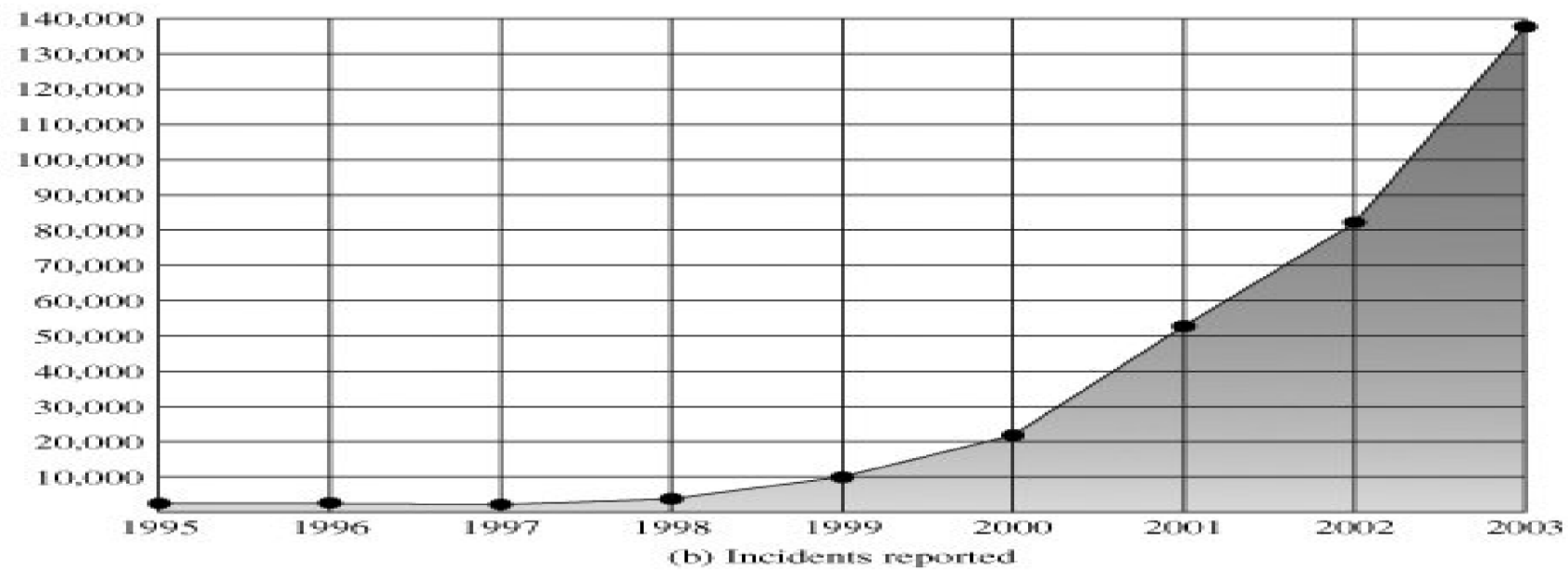
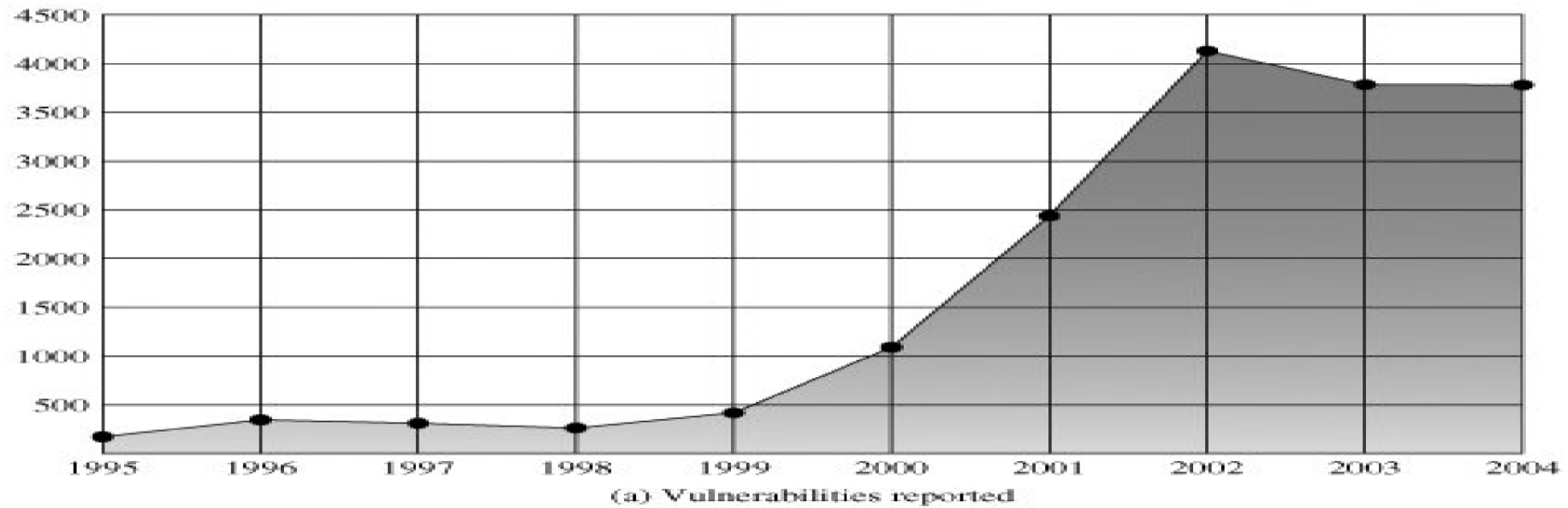
Figure 1.3 Model for Network Security

Methods of Defense

- Encryption
- Software Controls (access limitations in a data base, in operating system protect each user from other users)
- Hardware Controls (smartcard)
- Policies (frequent changes of passwords)
- Physical Controls

Example

- A security-related transformation on the information to be sent. Examples include the encryption of the message, which scrambles the message so that it is unreadable by the opponent, and the addition of a code based on the contents of the message, which can be used to verify the identity of the sender.
- Some secret information shared by the two principals and, it is hoped, unknown to the opponent. An example is an encryption key used in conjunction with the transformation to scramble the message before transmission and unscramble it on reception.

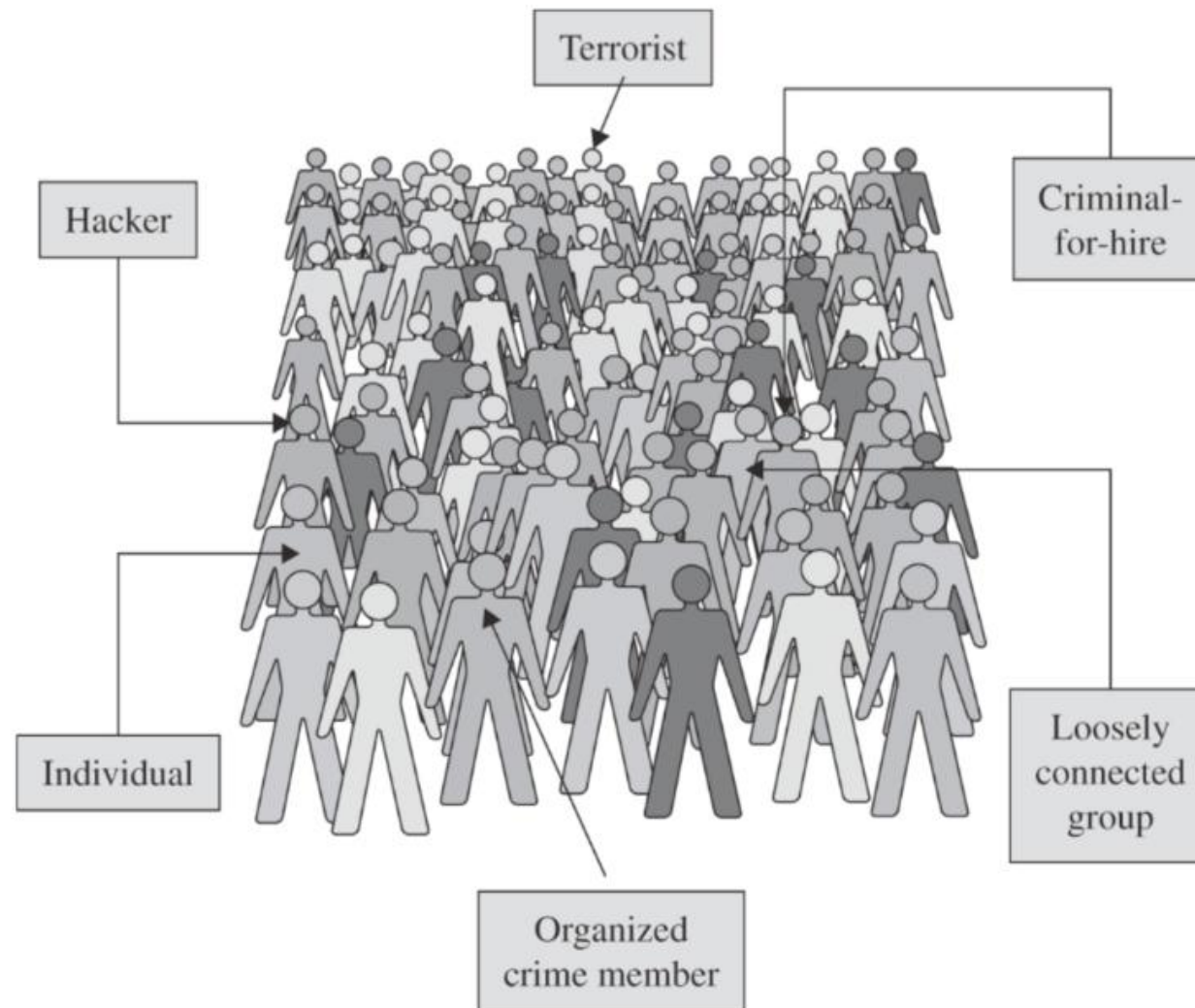


Secure Systems

•“The only truly secure system is one that is powered off, cast in a block of concrete and sealed in a lead-lined room with armed guards.” — Gene Spafford

- We have to make trade-offs between security and usability
- CIA: more ‘C’ usually means less ‘A’
- Security: a process, not a product
- We cannot make perfectly secure systems, but we can make our systems more defensible

Types of Attackers



Attacker Goals

- Why are our systems and networks attacked?
 - Steal our information or gather information
 - Steal our money
 - Use our hardware, software, or other assets
 - Destroy or deny use of our assets (data, information systems, physical resources)
 - Corrupt our information
 - Harm reputations, make a statement
 - Prepare for future action (eg botnets)
 - Just to see if it can be done
 - Penetration testing

Thank You